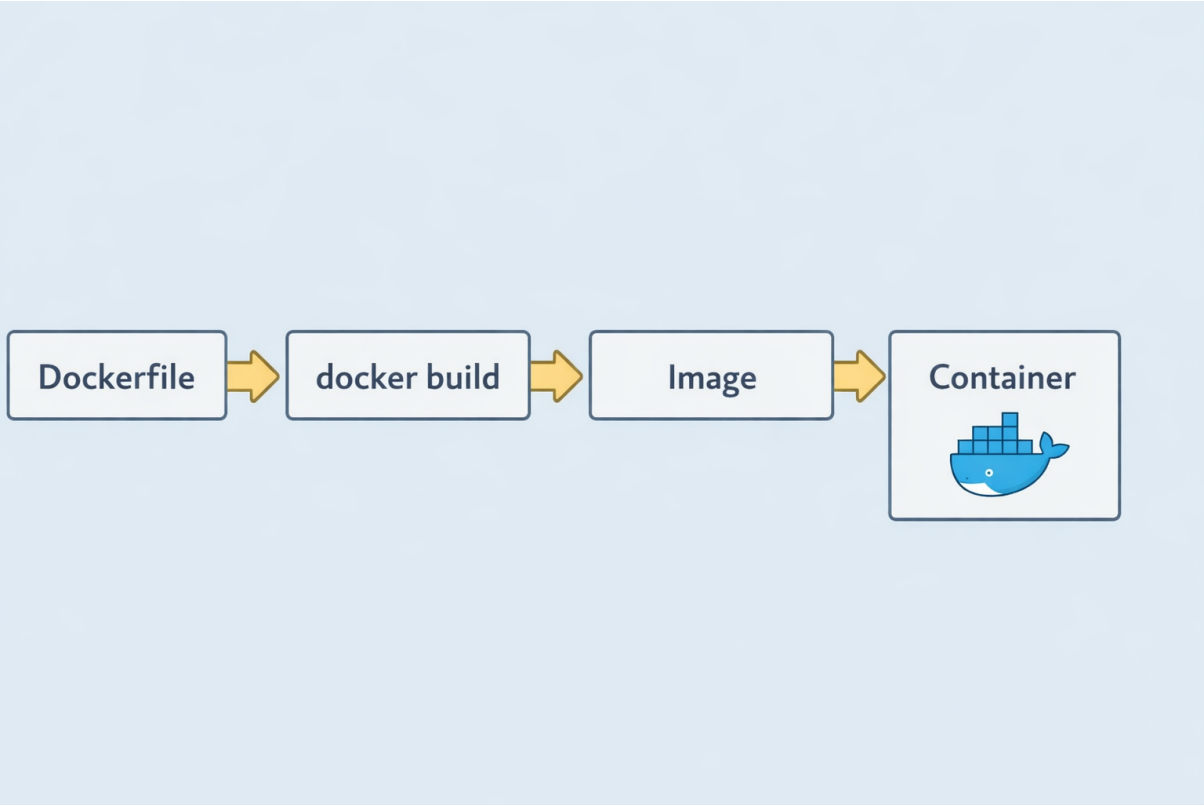
Dockerfile Creation:
What is a dockerfile?
A dockerfile is a text-based document that is used to create container images. It provides step by step instructions to create an image.

Key words:

```
Image = Blueprint (template)
Container = Running instance of image
```

Example:

```
docker build → creates IMAGE
docker run → creates CONTAINER
```



Key Dockerfile Instructions

Here are some key instructions:

Instruction	Purpose	Example
`FROM <image>`	Defines the base image to start the build process. Must be the first instruction.	`FROM ubuntu:latest`
`RUN <command>`	Executes any commands necessary to install packages or set up the environment *during* the image build.	`RUN apt-get update && apt-get install -y python3`
`CMD`	Provides defaults for executing a container. Only the *last* `CMD` instruction takes effect.	`CMD ["python3", "app.py"]`
`LABEL`	Adds metadata to an image (e.g., maintainer information).	`LABEL maintainer="user@example.com"`
`COPY <host-path> <image-path>`	Copies files or directories from the host machine (where the build is running) to the filesystem of the image.	`COPY . /app`
`ADD`	Similar to `COPY`, but can also handle remote URLs and automatically extract compressed files.	`ADD http://example.com/file.tar.gz /tmp/`
`EXPOSE`	Informs Docker that the container listens on the specified network ports at runtime. (It does not *publish* the port.)	`EXPOSE 8080`
`ENTRYPOINT`	Configures a container that will run as an executable. Often used with `CMD` to specify default arguments.	`ENTRYPOINT ["/usr/bin/nginx"]`
`WORKDIR <path>`	Sets the working directory for subsequent `RUN`, `CMD`, `ENTRYPOINT`, `COPY`, and `ADD` instructions.	`WORKDIR /app`
`ENV`	Sets environment variables. These variables are available to subsequent instructions and to the container at runtime.	`ENV PORT=8080`

Example 1:
We will make a simple image from a dockerfile. Hashtags # are for comments.
In this example we will take base layer ubuntu and install php inside it.

```
#Base Image
FROM ubuntu:latest
#Maintainer (optional)
LABEL maintainer="student@email.com"
#install Packages
RUN apt update && apt install -y php
#command when container runs
CMD ["echo", "Hello Students"]
```

Making new Directory using “mkdir”

Use any text editor (I’m using nano)

```
#base image
FROM ubuntu:latest

#maintainer (optional)
LABEL maintainer="Student Demo"

#install packages
```

```
LABEL maintainer="Student Demo"

#install packages
RUN apt update && apt install -y php

#default command when containers run
CMD ["echo", "Hello Students"]
```

Now we will build the image using command:

docker build .

docker build -t <image_name> .
-t: for tagging
“.” : path, “.” signifies current/present working directory

```
[+] Building 136.7s (6/6) FINISHED                                                                    docker:desktop-linux
 => [internal] load build definition from dockerfile                                                                  0.0s
 => => transferring dockerfile: 252B                                                                                  0.0s
 => [internal] load metadata for docker.io/library/ubuntu:latest                                                      0.4s
 => [internal] load .dockerignore                                                                                     0.0s
 => => transferring context: 2B                                                                                       0.0s
 => [1/2] FROM docker.io/library/ubuntu:latest@sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b      0.0s
 => => resolve docker.io/library/ubuntu:latest@sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b      0.0s
 => [2/2] RUN apt update && apt install -y php                                                                      132.0s
 => exporting to image                                                                                                4.2s
 => => exporting layers                                                                                               3.4s
 => => exporting manifest sha256:00ba525d1cbb37b631fd6d47a11f414458b30f57b06d6f84cc3d621606a091d9                     0.0s
 => => exporting config sha256:c38ad982b474ec5857d2a88523aaee6f810e1f013846dcf64cbd2364f58dde80                       0.0s
 => => exporting attestation manifest sha256:ebd4eed2f711bcfed0f436c203d21782e20858f6942589b634c647d3952ea894          0.0s
 => => exporting manifest list sha256:195916186bb26dd964bb38545c9c7e4d44f0a9a3f6ac298526e8c6a642621900                0.0s
 => => naming to docker.io/library/ulb:latest                                                                         0.0s
 => => unpacking to docker.io/library/ulb:latest                                                                      0.7s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/rz0mmrw9c6my7zbzgg2zrcf2d
sumitkumarmehta@sumits-MacBook-Air dockerfile % docker run ulb
Hello Students
sumitkumarmehta@sumits-MacBook-Air dockerfile % docker run -it ulb bash
root@8c31788e6a7b:/# php -v
PHP 8.3.6 (cli) (built: Jan  7 2026 08:40:32) (NTS)
Copyright (c) The PHP Group
Zend Engine v4.3.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.3.6, Copyright (c), by Zend Technologies
root@8c31788e6a7b:/# exit
exit
sumitkumarmehta@sumits-MacBook-Air dockerfile % docker images
                                                                                          i Info →   U  In Use
IMAGE                              ID             DISK USAGE   CONTENT SIZE   EXTRA
docker/welcome-to-docker:latest    c4d56c24da4f       22.9MB         6.35MB    U
httpd:latest                       dd178595edd6        207MB         47.8MB    U
mysql:latest                       932fe8fbc04c       1.28GB          278MB    U
nginx:latest                       13310a9cc1de        258MB         64.1MB    U
sakurawinter/ubuntusample:v1.1     067708b231d9        451MB          116MB
ubuntu:latest                      cd1dba651b30        141MB         30.8MB    U
ulb:latest                         195916186bb2        451MB          116MB    U
welcome-to-docker:latest           250e3657631c        441MB          122MB    U
```